2018A F=ma Exam Review

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January 30, 2026

Problem 4

To maximize the total E, you want to add the max KE, so the most effective place is perigee (same impulse, same Δv) because $\Delta K \propto v^2$, and ΔK would be larger if v was already large:

$$(100+5)^2 - 100^2 > (10+5)^2 - 10^2$$

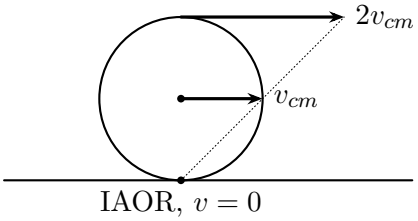
(an example where $\Delta v = 5$).

Problem 5

Consider the fact that there are no net external horizontal forces, so the com doesn't move horizontally.

Problem 8

Considering the IAOR to be the contact point, the top of the disk is always at $2v_{cm}$, because $v_{cm} - r\omega = 0$ for it to roll without slipping, so $v_{cm} = r\omega$, and the top of the disk is at $2v_{cm}$.



Problem 10

Think about the full picture. Identify what the graph should look like at the beginning, middle, and end, not just at one or two parts.

Problem 14

$$\frac{T_1}{T_2} = \frac{\omega_2}{\omega_1} = \frac{\sqrt{I_1}}{\sqrt{I_2}} = \frac{\sqrt{3m\ell^2}}{\sqrt{m\ell^2}} = \sqrt{3}.$$

mgd is the same for both ($3mg(L/3)$). ($d = L/3$ if you find x_{cm}, but it's actually unnecessary because mgd is the same for both and would cancel out. In these ratio problems you can often think in terms of proportionality.)

Problem 24

It initially increases because a larger-angle pendulum takes longer (by a factor of $1 + \theta^2/16 + \cdots$), but if you keep giving the mass a larger K, it'll swing in a circle, so of course the period of the motion would start decreasing once the circle gets completed at faster speeds.

Problem 25

- 1: $0.008/1.013 = 0.79\%$
- 2: $\sqrt{(0.008/1.013)^2 + (0.016/0.997)^2}\,/\,2 = 0.89\%$
- 3: $\sqrt{\left(4(0.008/1.013)\right)^2 + (0.016/0.997)^2}\,/\,5 = 0.71\%$

So $3 < 1 < 2$. Faster: Bob's period uncertainty is twice Alice's Δt, so Bob has $2\Delta t$ uncertainty, and you don't even need to plug in numbers:

- 1: Δt
- 2: $\sqrt{(\Delta t)^2 + (2\Delta t)^2}\,/\,2 = \frac{\sqrt{5}}{2}\Delta t = 1.1\,\Delta t$
- 3: $\sqrt{(4\Delta t)^2 + (2\Delta t)^2}\,/\,5 = \frac{2\sqrt{5}}{5}\Delta t = 0.9\,\Delta t$